Measuring Effective Data Resolution in Guided Diffusion Posteriors

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Abstract

Guided diffusion samplers are increasingly used to reconstruct physical fields from sparse observations, but standard diagnostics do not say how much of the reconstruction was actually determined by the data. We introduce effective data resolution for black-box generative posteriors: a comparison between the resolution warranted by the inverse problem, $\mathrm{dof}_{\mathrm{ref}}$, and the resolution realised by the sampler, $\mathrm{dof}_{\mathrm{samp}}$. A perturbation estimator measures $\mathrm{dof}_{\mathrm{samp}}$ and the spatial map $R(x, x)$ from sampler queries alone. We validate the estimator against exact references and use it to study guided diffusion. The resulting measurements show that guidance weight can strongly alter apparent information transfer, that mean, spread and resolution are not jointly corrected by one weight even with an exact prior and score, and that resolution fidelity does not follow reliably from the apparent principledness of a guidance rule.

1 Introduction

Reconstructing a physical field from a handful of sensors is a canonical ill-posed inverse problem [24, 25], and generative modelling has become a standard tool for it. The recipe is to train a diffusion prior over solution fields and then steer sampling toward the observations through a likelihood term, yielding full-field reconstructions with uncertainty from sparse data. DiffusionPDE [11], score-based inverse-problem samplers [22, 2, 14], score-based data assimilation [20] and function-space variants [26] are all instances of it.

Such a reconstruction is usually assessed by how close it lands to the truth and by whether its uncertainty is calibrated [9, 15]. These metrics do not, by themselves, identify the thing a scientist needs to know before drawing a conclusion from the reconstructed field: which parts of it the sensors actually determined. A sampler can produce accurate-looking, well-calibrated output while the structure a reader finds interesting is supplied almost entirely by the prior, or, in the other direction, while it commits to detail far beyond what the noise justifies [24]. Conventional diagnostics score the reconstruction and its marginals, not the dependence of the reconstruction on the data.

Two quantities, one comparison. The distinction we make precise is between the information the inverse problem warrants and the information the sampler realises.

The first is a property of the problem. Given a prior, a forward operator, a noise level and a sensor layout, there is a classical answer to how many directions of the unknown field the data can constrain:

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the resolution, or degrees-of-freedom-for-signal, of the inverse problem [1, 19, 24]. With n point sensors it is at most n, and in an ill-posed problem substantially less, because the forward operator destroys information before the sensors see it. It is computable in closed form from a Gauss–Newton Hessian without running any sampler. We write it $\mathrm{dof}_{\mathrm{ref}}$ and claim no novelty for it; we use it as a reference standard.

The second is a property of the algorithm. A guided sampler at a particular guidance weight implements some map from observations to reconstruction, $y \mapsto \mathbb{E}_{\mathrm{samp}}[u \mid y]$, and how strongly that map responds to the data is a fact about the sampler, not about the inverse problem. We call it the sampler's effective data resolution, $\mathrm{dof}_{\mathrm{samp}}$. It need not equal $\mathrm{dof}_{\mathrm{ref}}$, it is not currently reported, and, as we show, it is largely set by the guidance weight; Appendix D documents this for the DiffusionPDE configuration we inspected.

What we propose measuring is their ratio, the resolution-fidelity ratio

$$\rho_{\mathrm{res}} \;=\; \mathrm{dof}_{\mathrm{samp}}/\mathrm{dof}_{\mathrm{ref}}, \tag{1}$$

read as follows: near 1 the sampler attributes about as much resolution to the observations as they warrant; below 1 it attributes less, so the reconstruction leans more on the prior than it appears to; above 1 it attributes more than the noise justifies.

Why the sampler side is hard. For an estimator with an explicit Hessian, resolution is a matrix one differentiates. A guided diffusion posterior has neither a Hessian nor a closed form, can only be queried by running it, and is stochastic. The measurement therefore has to treat the sampler as a black box. We adapt the perturbation estimator for generalized degrees of freedom [28, 5], which perturbs the data and measures the response of the fit, combining it with stochastic trace estimation [12, 18]. The resulting estimator uses two evaluations of the sampler's reconstruction map per probe, a count that does not grow with the field dimension, and it also yields the spatial map. The scalar says how much information the data appear to have supplied; the map says where that apparent influence lands.

What the measurement reveals. We apply it to guided diffusion in two main settings and one appendix breadth check. In a 1D backward-heat problem with a Gaussian prior the diffused score and the true posterior are both available in closed form, so training error and prior mismatch are eliminated by construction and every deviation is attributable to the guidance scheme. There, the guidance weight moves effective resolution over 0.15 to 1.27 times the warranted value, and no single weight gets posterior mean, spread and resolution right together. In a 2D nonlinear Darcy problem with a non-Gaussian prior and a trained score, a comparable span reappears, but the advantage that principled guidance held in the controlled setting does not carry over. A non-PDE compressed-sensing check in Appendix E is included only as a modest breadth check: it confirms that the measurement is not tied to PDE forward operators, while leaving image restoration, MRI, CT and other application domains to future work.

Contributions.

1. **A measurement of effective data resolution.** We define $\mathrm{dof}_{\mathrm{samp}}$ for black-box generative posteriors, pair it with the classical reference $\mathrm{dof}_{\mathrm{ref}}$, and estimate the fidelity ratio ρ_{res} and spatial map $R(x, x)$ from sampler queries (§3, §4).
2. **Validation and operating conditions.** We certify the estimator against exact references (bias -0.009 ± 0.028, standard error $1/\sqrt{P}$), test it on a stochastic sampler, and identify the reproducibility, mixing and perturbation-scale conditions needed for trustworthy measurements (§5, §8).
3. **Empirical analysis of guided diffusion.** In a controlled 1D problem, guidance weight moves ρ_{res} over 0.15 to 1.27 and no weight is jointly correct for mean, spread and resolution; in nonlinear 2D Darcy, a comparable range appears but the controlled method ordering does not persist; a non-PDE compressed-sensing check gives a modest breadth test of the same separation between resolution and conventional metrics (§6, §7; Appendix E).

2 Related work

Guided diffusion for PDE inverse problems. DiffusionPDE [11] learns a joint distribution over coefficient and solution fields and guides sampling with sparse observations together with a PDE

residual. Score-based methods solve inverse problems by steering a learned prior at test time [22, 2, 14]; score-based data assimilation [20] decouples the observation model from an unconditional trajectory prior, and function-space formulations [26] target discretisation invariance. All of them steer an unconditional prior with an observation term, and all inherit an approximation in doing so, because the exact guidance term $\nabla_{x_t} \log p(y \mid x_t)$ is intractable in general. Diffusion posterior sampling (DPS) [2] replaces it with $\nabla_{x_t} \log p(y \mid \hat{x}_0(x_t))$, evaluating the likelihood at the denoised estimate. This substitutes a surrogate likelihood gradient for the intermediate-time likelihood gradient; its effect on how much information the sampler transfers has not, to our knowledge, been quantified. ΠGDM [21] models the $x_0 \mid x_t$ covariance isotropically, and DDNM [27] and DDRM [14] motivate data-consistency projection schemes. These form the approximation ladder that we use in the experiments.

Uncertainty quantification for neural PDE surrogates. Linearised neural operators yield function-valued Gaussian processes [17], and predictive calibration methods address reliability of reported intervals [15]; large pretrained PDE models [10] and neural operators [16] raise the same questions at scale. These quantify surrogate error: how wrong the learned operator is. Our question is different and complementary, concerning inverse-problem resolution, namely how much the sensors determined, which is a property of the data and the sampler rather than of the operator's accuracy.

Resolution analysis and perturbation-based degrees of freedom. The estimand is classical. Resolution operators and averaging kernels go back to Backus and Gilbert [1], degrees of freedom for signal are standard in atmospheric retrieval [19], and low-rank posterior structure is well studied in large-scale Bayesian inverse problems [25, 24, 7, 23]. So is the estimator: Ye [28] and Efron [5] define generalized degrees of freedom as $\sum_i \partial \hat{y}_i / \partial y_i$ and estimate it by perturbing the data and measuring the response of the fit. Monte-Carlo SURE uses the same black-box perturbation idea for denoisers [18], and Hutchinson probing [12] is the standard device for the trace.

Our contribution is not a new trace estimator. It is the transplant of this machinery to posteriors that can only be sampled, which requires common random numbers, a noise-scaled perturbation inside a window we characterise, and a smooth coupling condition; the certification of the resulting measurement against exact references; and the pairing of the measured quantity with a computable problem-side reference so that a fidelity ratio can be formed. To our knowledge no formulation of resolution analysis has been applied to a generative posterior, or used as a standard diagnostic for guided diffusion inverse problems.

3 Effective data resolution

Setting. Let $u \in \mathbb{R}^N$ be the unknown field on a discretised domain and let

$$y = Au + \eta \in \mathbb{R}^n, \qquad \eta \sim \mathcal{N}(0, \sigma_y^2 I), \qquad n \ll N, \tag{2}$$

be the observations, where A is the (linearised) forward operator mapping the field to sensor readings and σ_y is the noise level. An estimator is any map $m : \mathbb{R}^n \to \mathbb{R}^N$ from data to reconstruction; for us m will be the posterior mean of a sampler [24, 25].

The resolution operator, informally. Imagine changing the true field slightly at location x' and asking how the reconstruction at location x responds. Collecting these sensitivities gives the resolution operator

$$R \;=\; \frac{\partial m}{\partial u} \;=\; \frac{\partial m}{\partial y}\, A, \tag{3}$$

an $N \times N$ object whose (x, x') entry says how much a change in the truth at x' propagates into the estimate at x. A perfect estimator that recovered the field exactly would have $R = I$. In an ill-posed problem it cannot: the data do not determine the field, so R is a smoothing operator [1, 19].

Its diagonal $R(x, x)$ is the interpretable object. A value near 1 means the estimate at x tracks the truth at x, so that location is determined by the data. A value near 0 means the estimate is weakly determined by the data there, so prior information contributes more strongly to what appears at that location. The diagonal is thus a map of where data influence lands.

The scalar: effective data resolution. Summing the diagonal counts the number of independent directions the estimator lets the data determine,

$$\mathrm{dof}_{\mathrm{samp}} \;=\; \mathrm{tr}(R), \tag{4}$$

which we call the sampler's effective data resolution. It is a property of that algorithm at that guidance weight, not of the inverse problem: two samplers targeting the same posterior can, and do, return different values.

The reference: what the problem warrants. For the exact Bayesian posterior of a linear–Gaussian problem with prior covariance C, the corresponding quantity is classical and requires no sampling. The prior-preconditioned Gauss–Newton Hessian

$$H \;=\; C^{1/2} A^{\top} A\, C^{1/2} / \sigma_y^2 \tag{5}$$

has eigenvalues λ_i, each measuring how strongly the data constrain one prior-whitened direction, and

$$\mathrm{dof}_{\mathrm{ref}} \;=\; \sum_i \frac{\lambda_i}{1+\lambda_i}. \tag{6}$$

Each term lies in $[0, 1)$ and is near 1 for a well-constrained direction and near 0 for one the data do not see, so $\mathrm{dof}_{\mathrm{ref}}$ is the information budget the observations carry. Since $\mathrm{rank}(H) \le n$, it is cheap even at large N. For a nonlinear forward map we linearise at the posterior mean, which is standard practice in retrieval [19, 7, 23].

Resolution fidelity. The comparison we propose is the resolution-fidelity ratio $\rho_{\mathrm{res}} = \mathrm{dof}_{\mathrm{samp}}/\mathrm{dof}_{\mathrm{ref}}$ of (1). As a definition it says only this: $\rho_{\mathrm{res}} \approx 1$ means the sampler attributes about as much resolution to the observations as the inverse problem warrants, $\rho_{\mathrm{res}} < 1$ means it attributes less, and $\rho_{\mathrm{res}} > 1$ means it attributes more. Both departures occur in our experiments.

The interpretation is a separate step, and in our settings it is borne out by the other columns. A sampler with $\rho_{\mathrm{res}} < 1$ responds weakly to the data, so structure in its reconstruction is less data-driven than it appears and is correspondingly more attributable to the prior. A sampler with $\rho_{\mathrm{res}} > 1$ responds to the observations more strongly than the noise justifies, which in §6 coincides with variance ratios far above one and coverage at 100%: detail that the measurements do not support.

Relation to existing diagnostics. Coverage, CRPS [9] and posterior variance ratios score the distribution a sampler reports: whether its intervals contain the truth at the advertised rate, and whether its spread is right. Resolution is complementary rather than a replacement: it differentiates the reconstruction map with respect to y, and therefore measures data dependence that fixed-observation diagnostics do not directly access.

4 Black-box estimation

Computing R from (3) directly is an $N \times N$ object and out of reach. The reduction that makes the problem tractable is cyclicity of the trace,

$$\mathrm{tr}(R) \;=\; \mathrm{tr}\big(A\, \partial m/\partial y\big), \tag{7}$$

where $A\,\partial m/\partial y$ is only $n \times n$: the estimand lives in observation space, whose dimension is the sensor count, not in field space. We never form it. A Rademacher probe $z \in \{\pm 1\}^n$ gives the estimator

$$\mathrm{tr}(R) \;\approx\; \mathbb{E}_z\left[z^{\top} A\, \frac{m(y+\epsilon z) - m(y-\epsilon z)}{2\epsilon} \right], \tag{8}$$

averaged over P independent probes [12, 18]. Throughout, n denotes the sensor count and P the probe count; they are independent quantities.

Cost. Each probe requires two evaluations of the reconstruction map m, and the sampler is used only as a black box. It is worth being precise about what that costs, because $m(y)$ is itself the posterior mean of a stochastic sampler and must be approximated by an ensemble of E trajectories, so the total is $2PE$ trajectories. What does not grow with the field dimension N is the number of estimator evaluations, $2P$: the trace estimator introduces no dimensional dependence of its own, and whatever cost N imposes enters through the price of a single sampler trajectory, exactly as it would if one were merely drawing from the posterior.

The spatial map. Replacing the random probes by the n coordinate directions of y gives the columns of $\partial m/\partial y$ exactly, and hence the diagonal $R(x, x)$, at $2n$ evaluations and with no probe variance. This is affordable precisely in the sparse-sensor regime of interest, and it is what turns the scalar into the map of §3.

Common random numbers. The samplers we audit are stochastic, so $m(y)$ is estimated from a finite ensemble. The two evaluations in (8) must be run under common random numbers, that is, with the same seed, so that the ensemble noise cancels rather than being divided by 2ϵ. For a diffusion sampler this cancellation is exact rather than approximate: the noise draws depend only on the seed and never on y, so for a fixed seed $m(\cdot)$ is a deterministic function of the data [8].

Coupling requirement. The estimator therefore requires more than reproducibility at a fixed input: the two perturbed runs must remain smoothly coupled as y changes. Diffusion and flow samplers satisfy this. Metropolis-corrected samplers such as pCN [3] can be replayed from a seed, but their accept/reject discontinuities can split the two perturbed paths, so chain noise swamps the $O(\epsilon)$ signal. We return to this in §8.

Perturbation scale. The step ϵ must be scaled to the observation noise σ_y rather than set in absolute units, and it must lie inside a window. Its upper edge is ordinary finite-difference bias. It also has a lower edge for the residual-normalised DPS implementation we study, where the normalisation makes the reconstruction map near-singular at small scales; schemes without that normalisation show no such edge. We characterise the window in §8 and Appendix A, and report ϵ with every measurement.

5 Validation

Before using the measurement to draw conclusions about guided diffusion, we establish that the measurement itself is trustworthy. We do so on a backward-heat problem with a 128-dimensional field and 12 sensors in a half-domain layout, where the posterior is Gaussian in closed form and $\mathrm{dof}_{\mathrm{ref}} = 5.971$ exactly.

The estimator passes three checks (Table 3, Appendix C). First, it is unbiased against the exact reference: at $P = 128$ probes the bias is -0.009 ± 0.028. Second, its standard error follows the ordinary Monte-Carlo rate $1/\sqrt{P}$; increasing P from 8 to 128 lowers the per-run standard deviation $0.897 \to 0.179$, a factor 5.0 against the 4.0 predicted. Third, it works on a stochastic but reproducible sampler: preconditioned unadjusted Langevin gives 6.33 ± 0.59 against the exact 5.97.

This last check also fixes the main prerequisite. Resolution can only be measured from a sampler whose posterior mean is itself trustworthy. Without low-rank posterior preconditioning [7, 23], the unadjusted Langevin sampler [4] had effective sample size about 40 out of 150k steps and under-dispersed by 2 to $8\times$; after preconditioning it reached ESS $13{,}753$ and matched the closed-form posterior to 0.3% in mean and 0.6% in standard deviation. The perturbation scale is a separate condition: reference samplers with affine-in-y means are flat in ϵ, but guided samplers are not. A higher-replicate stress test in Appendix E confirms that the exact-score control remains stable while high-weight DPS becomes realisation-sensitive at reduced probe budgets. We return to these conditions in §8.

6 Guidance controls apparent information content

Design, and why it isolates guidance. We take a 1D backward-heat problem: a 64-dimensional field observed through 8 point sensors after diffusive smoothing, with $\mathrm{dof}_{\mathrm{ref}} = 7.85$. The prior is Gaussian, so the diffused score and the Tweedie denoiser [6] are available in closed form and the true posterior is known exactly. There is therefore zero training error and zero prior mismatch, and any departure from the reference posterior is attributable to the guidance scheme alone. This is what a released checkpoint cannot offer, since there guidance error, score error and prior mismatch are confounded. We compare exact guidance as a control arm against ΠGDM [21], a data-consistency projection in the style of DDNM [27], and DPS [2] at several guidance weights ζ. All resolution estimates use $\epsilon = \sigma_y$.

Table 1: Guidance weight trades posterior accuracy, spread, and effective resolution in the controlled 1D backward-heat problem. "mean err" and "std err" are relative L^2 errors in the posterior mean and pointwise standard deviation; "var ratio" is relative to the closed-form posterior, CRPS is the continuous ranked probability score, and "cov@90%" is empirical coverage of the nominal 90% interval. The fidelity ratio $\rho_{\text{res}} = \text{dof}_{\text{samp}}/\text{dof}_{\text{ref}}$ is measured at $\epsilon = \sigma_y$ over 4 ensemble realisations; $\pm$ denotes the across-realisation standard error.

guidance	mean err	std err	var ratio	CRPS	cov@90%	ρ_{res}
exact score (control)	0.040	0.046	0.97	0.309	0.812	**0.985** $\pm$**.002**
ΠGDM	0.041	0.047	0.96	0.307	0.812	1.022 $\pm$.002
projection	0.051	0.049	0.94	0.308	0.812	1.019 $\pm$.000
DPS $\zeta = 0.3$	0.155	0.521	2.22	0.314	0.938	0.796 $\pm$.038
DPS $\zeta = 1$	0.074	2.712	13.8	0.356	**1.000**	1.219 $\pm$.046
DPS $\zeta = 3$	0.350	9.453	97.7	0.528	**1.000**	1.053 $\pm$.129
DPS, literal $1/\sigma_y^2$	diverged ($\max \lvert x \rvert \sim 10^{173}$)					

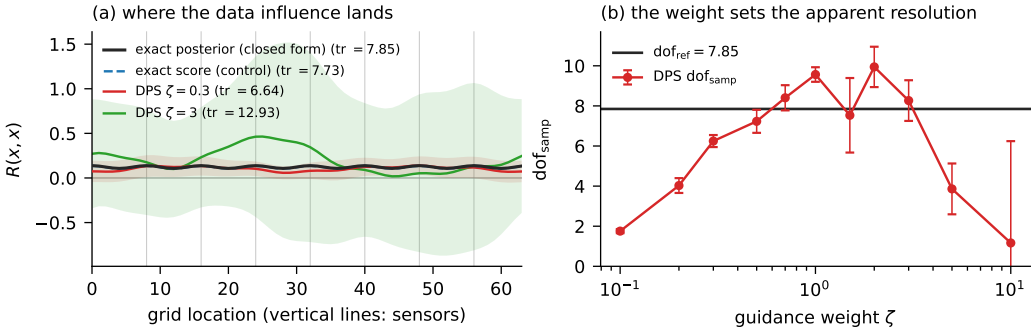


Figure 1: Effective data resolution in the controlled 1D setting, at $\epsilon = \sigma_y$. **(a)** The spatial map $R(x, x)$, averaged over 16 ensemble realisations with ± 1 sd bands, showing where apparent data influence lands. The exact posterior spreads it almost uniformly (0.107 to 0.138) and the exact-score control recovers that to 0.002 with zero spread across realisations, which validates the apparatus. DPS at $\zeta = 0.3$ under-resolves roughly uniformly (averaged trace 6.64 against 7.85), while DPS at $\zeta = 3$ concentrates influence in a band near $x = 24$ and starves the rest of the field (0.019 to 0.464); the latter's map is not even stable across realisations, its per-realisation trace ranging from -24.0 to 68.2. **(b)** The scalar dof_{samp} against guidance weight (4 realisations, error bars are across-realisation standard errors), with dof_{ref} as the horizontal line. The curve rises, crosses the reference, and falls back, so $\text{dof}_{\text{samp}} = \text{dof}_{\text{ref}}$ has more than one root and matching resolution does not identify a guidance weight.

The control arm validates the apparatus. Exact guidance returns $0.985\,\text{dof}_{\text{ref}}$, and ΠGDM and projection land within 2% (Table 1). Since these schemes closely match the reference posterior across the reported diagnostics, a departure elsewhere in the table is a property of the guidance scheme rather than of the measurement.

Guidance weight sets the apparent information content. Sweeping $\zeta \in [0.1, 10]$ (Figure 1b, Appendix D) moves dof_{samp} from 0.15 to 1.27 times dof_{ref}: at $\zeta = 0.1$ the sampler behaves as though the sensors had determined about a fifth of what they carry, near $\zeta = 2$ as though they had determined more than the noise justifies, and by $\zeta = 10$ it falls back to 0.15. The same observations, the same prior and the same score thus produce an eightfold difference in apparent information content, depending on a guidance weight rather than on the inverse problem alone. Thus, in this controlled setting, changing ζ substantially changes how much data dependence the sampler realises.

No single weight is jointly correct. Reading down the DPS rows of Table 1: the weight with the best posterior mean ($\zeta = 1$, mean error 0.074) carries $13.8\times$ the correct variance and reports 100% coverage at the nominal 90% level; the weight with the best distributional match ($\zeta = 0.3$) resolves only $0.80\times$ the warranted degrees of freedom; and the weight whose resolution is closest to correct

($\zeta = 3$, at $\rho_{\mathrm{res}} = 1.053$) has $97.7\times$ the correct variance and the worst CRPS in the table. No setting gets mean, spread and resolution right together. This is not a failure of tuning that a finer sweep would repair, because the sweep is what exhibits it, and it holds under the most favourable conditions available.

Mechanistically, the DPS rows differ from the exact and ΠGDM controls because the data term is evaluated at the denoised estimate and then scaled by the residual norm. Changing ζ therefore changes both the strength of the data pull and the region of the trajectory where residual-normalised corrections dominate the unconditional diffusion dynamics. Appendix A isolates the numerical consequence: for this DPS implementation the local derivative is singular as $\epsilon \to 0$, so the reported resolution is a finite-scale sensitivity, not an infinitesimal Jacobian.

Matching resolution is therefore not a tuning rule. Some ζ does hit $\mathrm{dof}_{\mathrm{ref}}$, but it is not the one with the best mean or spread, and the sweep is non-monotone, so $\mathrm{dof}_{\mathrm{samp}} = \mathrm{dof}_{\mathrm{ref}}$ has at least two roots, near $\zeta \approx 0.6$ and $\zeta \approx 4$, belonging to very different samplers. The measurement reports a discrepancy; it does not select a setting.

The scalar hides two spatial failures. Figure 1(a) shows what the trace alone cannot. DPS at $\zeta = 3$ mis-localises data influence: the exact posterior is nearly uniform over $[0.107, 0.138]$, whereas DPS spans $[0.019, 0.464]$ and concentrates influence near $x = 24$. The map is also unstable: across 16 ensemble realisations, $\mathrm{tr}(R)$ ranges from -24.0 to 68.2, whereas the exact-score control reproduces the closed-form map to 0.002 with zero spread. A sampler whose resolution map changes sign between runs is not one whose reported uncertainty can be taken at face value. Coverage and CRPS in Table 1 respond to the over-dispersion at $\zeta = 3$, but, being computed at fixed y, cannot reveal where the influence lands or whether it moves between runs.

What this experiment establishes, and what it leaves open. Because the score and the prior are exact here, this experiment establishes that effective resolution can change purely as a consequence of guidance, and within this controlled setting, distinguishes the schemes: exact guidance, ΠGDM and projection all recover $\mathrm{dof}_{\mathrm{ref}}$ to within 2%, while DPS does not produce a jointly correct posterior mean, spread and resolution at the weights studied. It cannot tell us whether either conclusion survives once the idealisations are removed. That is the next experiment.

7 Does resolution behaviour transfer? A nonlinear test

We therefore ask whether the behaviour of §6, and the relative standing of the guidance schemes within it, persist once score error, prior non-Gaussianity and forward-model nonlinearity are introduced together.

What becomes more realistic. We use a 2D Darcy flow problem: $24 \times 24 = 576$ field unknowns observed at 64 sensors, a nonlinear forward map $-\nabla \cdot (e^m \nabla u) = f$, a non-Gaussian prior $m = A \tanh(g/s)$ built over a latent Gaussian g (field kurtosis 1.88 against 3.0 for a Gaussian), and a trained 100k-parameter score network with EDM preconditioning [13]. Relative to §6 this adds nonlinearity, prior non-Gaussianity and score approximation simultaneously.

An exact reference survives. The latent construction is what keeps the comparison meaningful. Although the field prior is non-Gaussian, it is a deterministic transform of a Gaussian latent, and preconditioned Crank–Nicolson [3] in that latent space is Metropolis-corrected and therefore exact for the true posterior. This gives a trustworthy reference ensemble; $\mathrm{dof}_{\mathrm{ref}} = 43.15$, i.e. 7.5% of the 576 field unknowns. Resolution estimates use $P = 24$ Rademacher probes and $\epsilon = 0.5\,\sigma_y$.

The central phenomenon replicates: across DPS weights, ρ_{res} ranges from 0.15 to 1.37, comparable to the 0.15 to 1.27 of §6. What does not transfer is the relative standing of the guidance schemes. Projection was within 2% of the warranted resolution in the controlled setting, but here over-resolves by 50%; DPS at $\zeta = 1$ is the closest listed method at $\rho_{\mathrm{res}} = 1.026$. Nor does over-dispersion transfer in the same form: coverage stays between 0.785 and 0.851 against the reference's 0.774, and spread ratios are 1.2–1.6 rather than 13.8–97.7.

The scope is deliberately narrow. ΠGDM diverged in our nonlinear implementation, so projection is the only principled scheme represented here; we therefore claim only that projection does not

Table 2: Resolution fidelity under a nonlinear map with a trained score, against an exact pCN reference. "rel-L^2" is the relative L^2 error of the posterior mean, "std/ref" is ensemble spread relative to the pCN reference, "SW→ref" is the sliced-Wasserstein distance to that reference, CRPS is the continuous ranked probability score, and "cov@90%" is empirical coverage of the nominal 90% posterior interval. The span of ρ_{res} is comparable to the controlled setting, but projection over-resolves by 50% while DPS at $\zeta = 1$ is the closest listed method. ΠGDM diverged in our implementation, so we exclude it rather than claim a result about the method.

guidance	rel-L^2	std/ref	SW→ref	CRPS	cov@90%	ρ_{res}
pCN (reference)	0.689	1.000	0.000	0.211	0.774	1.000
ΠGDM	diverged (implementation; see text)					
projection	0.808	1.219	0.255	0.249	0.785	**1.501**
DPS $\zeta = 0.3$	0.911	1.644	0.341	0.283	0.851	0.152
DPS $\zeta = 1$	0.796	1.373	0.270	0.244	0.845	**1.026**
DPS $\zeta = 3$	0.799	1.290	0.252	0.245	0.812	1.372

retain its 1D advantage, and that a scheme's approximation structure did not predict ρ_{res} across these settings. The projection result is not caused by the stale-Jacobian approximation: refreshing $JJ^\top$ every $25, 5, 1$ steps gives $\rho_{\mathrm{res}} = 1.501, 1.482, 1.479$.

8 Use and scope

The measurement is useful only when its operating conditions are made explicit. The perturbation scale must be tied to the observation noise. In 2D Darcy, using the 1D absolute value $\epsilon = 10^{-2}$ perturbs the data by $30\times$ the noise and returns $\mathrm{dof}_{\mathrm{samp}} = 10.0$ against a true 43.1. Nor can ϵ always go to zero: for the residual-normalised DPS implementation studied here, $\|y - A\hat{x}_0\|^{-2}$ makes the small-scale derivative singular (Appendix A). We therefore report ϵ, use $\epsilon \in [0.1, 1]\sigma_y$, and make no claim about DPS formulations with different scalings.

The estimator also requires a reproducible, smoothly coupled sampler under common random numbers. Diffusion samplers satisfy this; Metropolis-corrected samplers such as pCN [3] can be replayed from a seed, but accept/reject discontinuities can split the two perturbed paths. For pCN we use the Gauss–Newton reference (6), not black-box $\mathrm{dof}_{\mathrm{samp}}$. Randomised traces are useful mainly when observation dimension is large: the coordinate-wise map costs $2n$ evaluations and has no probe variance, while our calibration suggests $P \approx 12$ probes for 10% accuracy and $P \approx 289$ for 2%.

Our scale is deliberate: the comparison requires a trustworthy $\mathrm{dof}_{\mathrm{ref}}$, available analytically in 1D and through latent pCN in 2D. Released checkpoints, image restoration, MRI, CT, robotics, consistency models, flow matching, rectified flows and newer inverse solvers are natural targets, but each requires a defensible reference resolution. Our claims are therefore limited to the guidance schemes implemented here, with nonlinear resolution interpreted as a posterior-mean linearisation and severe over-dispersion treated as a practical measurability limit. Appendix E reports a non-PDE compressed-sensing check (Table 4) and a higher-replicate stability stress test (Table 5).

9 Conclusion

A generative posterior can produce accurate reconstructions and plausible uncertainty statistics while transferring the wrong amount of information from the data. We made that quantity measurable by comparing $\mathrm{dof}_{\mathrm{ref}}$, the problem-side resolution budget, with $\mathrm{dof}_{\mathrm{samp}}$, the sampler-realised resolution, and by estimating both ρ_{res} and the map $R(x, x)$ from black-box sampler queries. Applied to guided diffusion, the measurement shows that guidance weight can move apparent information content by nearly an order of magnitude, that no single weight fixes mean, spread and resolution together even with an exact score and prior, and that the controlled method ordering need not persist under nonlinear learned settings. Information fidelity is therefore something to measure, not something to infer from the form of a guidance derivation.

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A The perturbation scale: window and singular behaviour

Why the certification of §5 does not generalise. The estimate is flat across $\epsilon \in [10^{-5}, 10^{-1}]$ to twelve significant figures on both reference samplers. Both have a posterior mean that is affine in y at a fixed noise realisation: the exact Gaussian posterior by construction, and preconditioned unadjusted Langevin because its update is affine in y at fixed noise. For an affine map the central difference is exact at every ϵ, so flatness there is guaranteed and carries no information about guided samplers.

Common random numbers are exact, so this is bias and not noise. The sampler draws its noise from a generator seeded independently of y, so for a fixed seed $\hat{m}(y)$ is a deterministic, smooth function of y and the two evaluations in (8) share their randomness exactly. There is no Monte-Carlo residual being amplified by $1/2\epsilon$. What varies between runs is which realisation $\hat{m}$ one obtains, which is why we report across-realisation error bars throughout.

Residual-normalised DPS admits no $\epsilon \to 0$ limit. DPS Algorithm 1 divides its correction by $\|y - A\hat{x}_0\|^2$, giving $\hat{m}$ near-singular structure at small scales. The statements below concern that implementation; other scalings of the guidance step may behave differently and we did not test them. Measuring $\zeta = 1$ with ensembles of 512:

ϵ/σ_y	10^{-6}	10^{-4}	10^{-2}	0.1	1	10
$\mathrm{dof}_{\mathrm{samp}}$	-5.25×10^6	4.27×10^4	462.2	10.94	10.91	8.35

Differentiating the sampler confirms it. The map is differentiable in y by automatic differentiation, so $\mathrm{tr}(R) = \mathrm{tr}(A\,\partial\hat{m}/\partial y)$ follows from n reverse-mode passes with cotangent $A^\top e_k$, at no ϵ at

all. On schemes without residual normalisation this reproduces the finite difference exactly and is independent of the realisation: exact score 7.733, ΠGDM 8.016, projection 8.000, each with zero spread across seeds and across ensemble sizes from 128 to 2048. On every DPS setting it returns values of order 10^{47}, which is the correct answer to a question that should not be asked of that sampler.

Choosing inside the window. The upper edge is ordinary secant bias and the lower edge is the singularity above. Across four realisations in 1D, the bias–variance tradeoff places the operating point at $\epsilon = \sigma_y$: at $0.1\,\sigma_y$ the estimate is essentially unbiased but the across-realisation standard deviation is 8 to 100, while at $10\,\sigma_y$ the spread is small but the value is biased low (5.54, 7.66, 7.88 for $\zeta = 0.3, 1, 3$ against 6.25, 9.57, 8.27 at σ_y). All 1D results use $\epsilon = \sigma_y$. All 2D results use $\epsilon = 0.5\,\sigma_y$, where the low-ζ estimate is stable (6.650 at $0.1\,\sigma_y$ against 6.578 at $0.5\,\sigma_y$ for $\zeta = 0.3$); at larger ζ the 2D estimates retain some probe-scale sensitivity, so the 2D values in Table 2 should be read as accurate to their leading digit rather than to three.

B Three formulations that do not work

Three natural-looking alternatives to $\mathrm{tr}(R)$ are degenerate, one of them subtly. We record them so that they are not re-proposed.

(i) The zero-prior limit. Sweeping prior strength and calling $\lim_{\lambda\to 0} \mathrm{Var}[u \mid y]$ the irreducible ambiguity. With sparse sensors this limit diverges, so the intended difference of variances is a difference of infinities.

(ii) Fraction of prior variance removed. $\rho(x) = 1 - \mathrm{Var}_{\mathrm{post}}(x)/C(x,x)$. For a stationary prior $C(x,x)$ is constant, so ρ is an affine function of the posterior variance map and carries no information the variance map does not already have. Our two-panel version of this diagnostic, which displayed the two as though they were independent views, had panels agreeing to four decimal places.

(iii) Elasticity under prior rescaling. $f(x) = \partial \log \mathrm{Var}_{\mathrm{post}}(x)/\partial \log \kappa^2$ for a prior scaled by κ^2. This one is subtle, because it is well defined and we verified it to 1.7×10^{-4} against finite differences before noticing the problem. In the eigenbasis of the prior-preconditioned Hessian it weights direction i by $1/(1+\lambda_i)$, which is near zero exactly where λ_i is large, that is, exactly in the well-informed directions one is trying to see; at $\lambda \approx 4 \times 10^4$ the weight is 2×10^{-5}. On a planted test with a deliberately inhomogeneous information distribution it produced 1.1% spatial contrast. It measures where the prior still dominates, which is the complement of the question.

C Certification details

Backward heat with $N = 128$, $\nu = 5 \times 10^{-3}$, $t_{\mathrm{obs}} = 1$, 12 sensors in a half-domain layout, $\sigma_y = 10^{-2}$, and prior parameters $\alpha = 1$, $\tau = 8$. The posterior is Gaussian in closed form, giving $\mathrm{dof}_{\mathrm{ref}} = 5.971$ exactly. Table 3 reports unbiasedness and the convergence rate.

Table 3: Certification of the black-box estimator. Backward heat, $N = 128$, $n = 12$, exact posterior, $\mathrm{dof}_{\mathrm{ref}} = 5.971$; 40 independent probe sets per row. From $P = 8$ to $P = 128$ the per-run standard deviation falls by a factor 5.0 against the 4.0 predicted by $1/\sqrt{P}$.

P (probes)	solves	bias	sd of mean	per-run sd
8	17	-0.094	0.142	0.897
32	65	-0.011	0.070	0.442
128	257	-0.009	0.028	0.179

From $P = 8$ to $P = 128$, a $16\times$ increase, the per-run standard deviation falls $0.897 \to 0.179$ and the standard deviation of the mean falls $0.142 \to 0.028$, both factors of 5.0 against the 4.0 predicted by $1/\sqrt{P}$. On the stochastic sampler, preconditioned unadjusted Langevin returns 6.33 ± 0.59 against the exact 5.97.

Sampling infrastructure. Prior-preconditioned sampling gave an effective sample size of about 40 out of 150k steps with under-dispersion of 2 to $8\times$ across all schemes, exact ones included. Low-rank posterior preconditioning [7, 23] at rank equal to the sensor count, which is exact because point data inform at most n directions, raised the unadjusted-Langevin stability threshold from 9.7×10^{-5} to 2.0 and the effective sample size from 40 to $13{,}753$, after which the sampler matched the closed-form posterior to 0.3% in mean and 0.6% in standard deviation.

D Full guidance-weight sweep

The sweep behind Figure 1(b). Backward heat with 64 field unknowns and 8 sensors, $\mathrm{dof}_{\mathrm{ref}} = 7.848$, $\epsilon = \sigma_y$, $P = 24$ probes, ensembles of 512, averaged over 4 realisations with $\pm$ the across-realisation standard error.

ζ	$\mathrm{dof}_{\mathrm{samp}}$	stderr	ρ_{res}
0.1	1.755	0.103	0.224
0.2	4.028	0.370	0.513
0.3	6.247	0.300	0.796
0.5	7.231	0.575	0.921
0.7	8.405	0.628	1.071
1	9.565	0.363	1.219
1.5	7.532	1.859	0.960
2	9.945	1.007	**1.267**
3	8.266	1.015	1.053
5	3.861	1.267	0.492
10	1.165	5.077	**0.148**

Two features matter. The curve is not monotone in ζ: it rises to a peak near $\zeta = 2$ and falls back, so $\mathrm{dof}_{\mathrm{samp}} = \mathrm{dof}_{\mathrm{ref}}$ has at least two roots and matching resolution does not identify a weight. And the estimate degrades at the top end, where the ensemble is over-dispersed by two orders of magnitude (Table 1) and the across-realisation spread grows accordingly; the $\zeta = 10$ entry has a standard error larger than its value. We therefore quote the range as a ratio, 0.15 to 1.27 times $\mathrm{dof}_{\mathrm{ref}}$, rather than as an endpoint-to-endpoint multiplier, which would rest on the noisiest entry in the table.

Guidance weights are real configuration choices. The weights swept here are not a strawman. DiffusionPDE's shipped Darcy configuration uses three hand-tuned guidance weights spanning three orders of magnitude, differing per PDE family and absent from that paper's result tables [11].

E Additional breadth and stability checks

A non-PDE compressed-sensing inverse problem. To check that the measurement is not tied to PDE forward maps, we repeated the controlled experiment with random compressed linear observations, $y = Au + \eta$, where $u \in \mathbb{R}^{64}$ has the same exact Gaussian prior, $A \in \mathbb{R}^{16\times 64}$ has independent Gaussian rows normalised to unit norm, and $\sigma_y = 0.05$. This is not a new modelling benchmark; it is a breadth check in which the prior, diffused score, denoiser and posterior reference all remain exact. The warranted resolution is $\mathrm{dof}_{\mathrm{ref}} = 15.78$, close to the sensor count because the random measurements are informative under this prior.

The compressed-sensing result is deliberately modest and should not be read as a third application benchmark. It does not address released checkpoints, independent codebases or image-domain inverse solvers, but it does remove the most immediate concern that the measurement only behaves on PDE operators. The same pattern recurs: exact guidance is certified against the reference, and ρ_{res} provides information not contained in CRPS, coverage or posterior variance.

Higher-replicate stability stress test. We also repeated the scalar 1D measurement for the rows that drive Table 1, using 6 independent realisations and $P = 12$ probes per realisation. This is intentionally a stress test with half the probe budget of the main table, so the numbers below are not used as headline estimates. They confirm the control arm and reveal the finite-budget failure mode:

Table 4: Non-PDE breadth check: random compressed sensing with an exact prior, exact score and exact posterior reference. Resolution is estimated with $\epsilon = \sigma_y$, $P = 24$ probes, ensembles of 512, and 4 independent realisations. The exact-score control again recovers $\mathrm{dof}_{\mathrm{ref}}$, while distributional spread and resolution fidelity separate: projection has nearly correct ρ_{res} but about $2.1\times$ posterior variance, and DPS can have near-reference ρ_{res} with much larger spread.

guidance	mean err	var ratio	CRPS	cov@90%	ρ_{res}
exact score	0.102	0.96	0.271	0.812	0.991 $\pm$.001
ΠGDM	0.246	1.17	0.283	0.828	0.899 $\pm$.004
projection	0.603	2.11	0.484	0.797	1.014 $\pm$.000
DPS $\zeta = 0.3$	0.697	21.4	0.521	0.922	1.023 $\pm$.064
DPS $\zeta = 1$	0.409	17.0	0.402	1.000	1.079 $\pm$.091
DPS $\zeta = 3$	0.126	33.4	0.393	1.000	0.839 $\pm$.066

once DPS becomes highly over-dispersed, across-realisation variation dominates the probe standard error.

Table 5: Higher-replicate 1D stability stress test, with $\mathrm{dof}_{\mathrm{ref}} = 7.85$, $\epsilon = \sigma_y$, $P = 12$ probes, ensembles of 512, and 6 realisations. The exact control remains stable. Low-weight DPS remains below the warranted resolution, while larger weights are noisy at this reduced probe budget, consistent with the unstable high-ζ maps in Figure 1.

guidance	$\mathrm{dof}_{\mathrm{samp}}$	ρ_{res}	sd(ρ_{res})	median probe se
exact score	7.77 ± 0.03	0.990	0.009	0.039
DPS $\zeta = 0.3$	5.50 ± 0.39	0.701	0.123	1.128
DPS $\zeta = 1$	8.92 ± 1.94	1.137	0.606	3.743
DPS $\zeta = 3$	6.23 ± 4.32	0.793	1.347	10.282

The important point is not the shifted point estimates at reduced P, but the scale of the uncertainty. Exact guidance remains repeatable. DPS at large guidance weights does not: the same row that gives $97.7\times$ posterior variance in Table 1 has realisation values from -9.94 to 17.21 in this stress test. This supports the paper's practical recommendation to report perturbation scale, probe budget and realisation variability rather than a single scalar alone.